A E G I S

A-E-G-I-S HEXAPOD ROBOT

Introduction

Emergency response operations can require people to enter environments that are unstable, hazardous, difficult to access or poorly understood. These conditions may occur after structural collapses, fires, industrial accidents, mining incidents, earthquakes, landslides and other emergencies. In such situations, responders may need to locate victims, identify hazards and understand the surrounding environment while operating under significant time and safety pressures.

The National Institute of Standards and Technology (NIST) identifies emergency response as an environment in which responders can face extreme hazards. Its research on emergency response robots specifically identifies applications such as searching for survivors in collapsed or compromised structures, establishing situational awareness around fires, and assessing large industrial or transportation accidents. NIST therefore recognises remotely operated robots as a potential means of allowing responders to obtain information while reducing their exposure to dangerous environments.

The challenge is particularly important when the environment contains rubble, uneven terrain, confined passages, smoke, poor visibility or unstable structures. A robot operating remotely can potentially enter an area before a human responder and provide information about the environment. NIST describes response robots as tools that can act as the “eyes and ears” of responders in dangerous, unknown or difficult-to-access locations.

The problem is also relevant to South African industries such as mining. According to the South African Department of Mineral Resources and Petroleum, the mining industry recorded 42 occupational fatalities in 2024. Although this represented the lowest number of mining fatalities recorded in the country's history and a 24% improvement from 2023, the same report recorded 1,841 occupational injuries during 2024. The statistics demonstrate that hazardous working environments remain an important safety concern.

Against this background, the Mini Search and Rescue Hexapod Robot is proposed as a small robotic platform capable of remotely entering difficult environments and collecting information through onboard sensing and vision systems. The project focuses on using legged mobility to investigate environments where conventional access may be difficult, while allowing human operators to remain at a safer distance.

Problem Statement

During emergencies and hazardous industrial operations, responders and workers may need to enter environments before having sufficient information about the conditions inside. Collapsed structures, rubble, confined spaces, unstable ground, industrial hazards and limited visibility can make direct human access dangerous.

The central problem addressed by this project is therefore:

Emergency responders and workers may lack a safe, mobile and accessible method of obtaining real-time information from hazardous or difficult-to-access environments before entering them.

Current response activities often require humans to assess conditions, locate victims or inspect potentially dangerous areas. However, the environment itself may expose personnel to additional risks. NIST research identifies the need for robotic systems that can operate remotely, provide situational awareness and perform tasks such as mobility, sensing, communication and searching in complex environments.

This creates an opportunity for a compact robotic platform that can enter difficult areas, collect environmental information and transmit that information to a human operator. The purpose is not to replace emergency personnel, but to provide them with additional information before or during intervention.

The Mini Search and Rescue Hexapod Robot addresses this problem by combining:

- Six-legged mobility for movement across irregular terrain.
- Remote operation to keep the human operator away from immediate hazards.

- Cameras and sensors for environmental observation.
- Computational control for coordinated leg movement.
- A compact structure designed for potentially confined environments.
- Real-time information gathering to support human decision-making.

Who is affected?

The problem affects several groups of users and stakeholders.

1. Emergency responders

Firefighters, urban search-and-rescue teams and other emergency personnel may be required to investigate dangerous environments while searching for victims. NIST identifies search-and-rescue operations in collapsed or compromised structures as situations where response robots can assist responders and reduce exposure to hazards.

2. Mine rescue and mining personnel.

Mining personnel can encounter hazards associated with unstable ground, machinery, transportation systems and confined underground environments. South African government statistics recorded 42 mining fatalities and 1,841 occupational injuries in 2024, demonstrating the continued importance of mine safety.

3. Industrial inspection teams.

Workers responsible for inspecting industrial facilities may need to access areas that are hazardous, confined or difficult to reach. A remotely operated robot could potentially collect visual or sensor information before personnel enter the area.

4. Disaster-management organisations.

Organisations responsible for disaster response may require rapid information about damaged buildings, blocked areas, unstable structures and potential victims before deciding how to deploy personnel.

5. The general public and victims.

Although victims may not operate the robot themselves, they are important beneficiaries. Faster information gathering and safer access could assist responders in making better-informed decisions when searching for people in hazardous environments.

The primary user of the proposed system is therefore expected to be a trained emergency or inspection operator, while the wider beneficiaries include emergency teams, organisations, workers and members of the public who may be affected by hazardous incidents.

3.4 Problem Validation.

The problem is supported by evidence from established research organisations and South African occupational safety statistics.

Evidence 1: Emergency responders operate in hazardous environments

The National Institute of Standards and Technology (NIST) states that emergency responders can risk their lives while dealing with extreme hazards. NIST specifically identifies searching for survivors in collapsed or compromised structures, establishing situational awareness around fires and assessing large industrial or transportation accidents as applications for emergency response robots.

This provides evidence that the need for robotic assistance in hazardous environments is not only theoretical. Government-supported research is actively investigating robotic systems for these types of emergency missions.

Evidence 2: Robots are being developed specifically for search and rescue.

NIST and the U.S. Department of Homeland Security have developed performance standards and testing methods for urban search-and-rescue robots. These tests evaluate capabilities including mobility, sensing, navigation, communication, operator control and safety.

NIST explains that these robotic systems are intended to help responders operate in extremely challenging conditions and have the potential to take responders out of harm's way while augmenting their capabilities.

This validates the fundamental concept behind the project: robots can be used as an additional tool to obtain information and perform tasks in environments where human access is dangerous.

Evidence 3: Difficult terrain and confined spaces are recognised robotic challenges.

NIST research on mobile-robot performance identifies the ability to negotiate different terrain types, obstacles and confined passageways as important requirements for robots used in critical operations.

This is directly relevant to the proposed hexapod design because the project is specifically investigating a legged platform for movement across irregular and constrained environments.

Evidence 4: South African mining provides local statistical evidence

The problem also has South African relevance. According to the South African Department of Mineral Resources and Petroleum's 2024 Mine Health and Safety Statistics:

- 42 occupational fatalities were recorded in South African mining in 2024.
- This represented a 24% improvement from 55 fatalities in 2023.
- 1,841 occupational injuries were reported in 2024.
- 13 fatalities in 2024 were associated with fall-of-ground accidents.

Although the statistics show significant improvement in mine safety, they also demonstrate that hazardous working environments continue to result in serious injuries and loss of life.

Evidence 5: The problem extends beyond mining.

The evidence from NIST demonstrates that hazardous-access problems occur across multiple environments, including collapsed structures, fires, industrial accidents and hazardous-material incidents. NIST describes robots as potentially providing situational awareness from a safer distance and helping responders investigate environments that are dangerous or difficult to access.

Therefore, the problem is not limited to one industry. The same underlying need exists across emergency response, mining, industrial inspection, disaster management and other hazardous-environment applications.

Validation conclusion.

The available research and statistical evidence support the existence of a genuine problem: people are required to obtain information and perform tasks in environments where direct human access can expose them to significant hazards.

Existing research by NIST demonstrates that robotic systems are already being investigated and evaluated for these conditions, while South African occupational safety statistics demonstrate the continued relevance of hazardous environments locally.

The proposed Mini Search and Rescue Hexapod Robot therefore addresses an established need rather than creating a problem solely for the purpose of developing a robot. Its proposed contribution is to investigate whether a compact, legged, remotely operated platform can provide a practical and potentially affordable method of accessing difficult environments and collecting information for human decision-makers.

Proposed Solution

The proposed solution is the Robo Rescue Spider, a compact six-legged robotic platform designed to support emergency responders and personnel operating in hazardous or difficult-to-access environments.

The system is designed to provide remote access, environmental observation and situational awareness without requiring a human operator to immediately enter the potentially dangerous area. The robot carries a camera and other sensors that can collect information about the surrounding environment and transmit it to a remote operator.

The concept is based on the use of a mobile robotic platform as an additional tool for human responders rather than as a replacement for them. This approach is consistent with research conducted by the National Institute of Standards and Technology (NIST), which identifies emergency-response robots as systems that can help responders obtain information and perform tasks from safer distances in hazardous environments .

The Robo Rescue Spider combines four main elements:

Legged mobility – six independently controlled legs provide movement across uneven terrain and obstacles.

Sensing and vision – onboard cameras and sensors provide information about the surrounding environment.

Remote operation – a human operator controls and monitors the robot from a safer location.

Computational control – mathematical models, inverse kinematics and gait algorithms coordinate the movement of the six legs.

The proposed system is intended to be adaptable to different applications, including search and rescue, mine inspection, industrial inspection, disaster assessment and other situations where direct human access may be hazardous.

The solution therefore addresses the identified problem by creating a robotic “first look” capability: the robot enters, observes and reports information before or while humans make decisions about entering the environment.

4.2 How Robo Rescue Spider Works

The Robo Rescue Spider operates through an interaction between the operator, control system, robotic body and sensing system.

Step 1: Deployment

The robot is placed near the area that requires investigation. Depending on the application, it can be deployed manually by a responder or positioned at the entrance of a hazardous or confined environment.

Step 2: Remote control

A human operator sends movement commands to the robot through a control interface. The operator can command movements such as forward, backward, turning and stopping.

Remote operation is important because it allows the operator to remain outside the immediate hazardous environment. NIST identifies safe-distance operation and reliable communication as important considerations for emergency-response robots .

Step 3: Movement and gait control

The robot converts the operator's movement commands into coordinated movements of its six legs.

The control system determines the position of each leg using mathematical models and inverse kinematics. The system then generates appropriate leg trajectories and gait patterns so that the robot can move while maintaining stability.

A six-legged robot can use different gait patterns depending on the required movement and terrain. For example, a tripod gait can coordinate three legs in contact with the ground while the other three move, before alternating the groups.

Step 4: Environmental sensing

While moving, the robot uses its camera and sensors to observe the surrounding environment.

The sensing system can provide information such as:

- Visual conditions.
- Obstacles.
- Possible pathways.
- Environmental conditions.
- Potential hazards.
- Possible locations requiring further investigation.

NIST research identifies sensing, mobility, communication and situational awareness as important capabilities for robots used in emergency-response applications .

Step 5: Information transmission.

Information collected by the robot is transmitted to the operator.

The operator can therefore observe the environment without physically entering it. This creates a human–robot system in which the robot performs remote observation while the human remains responsible for interpretation and decision-making.

Step 6: Human decision-making.

The information collected by the robot can assist responders in deciding what action should be taken next.

For example, the operator may determine that:

- An area is inaccessible.
- A pathway is available.
- An obstacle needs to be avoided.
- A location requires further investigation.
- Human personnel should not enter the area immediately.

The robot therefore functions as an information-gathering and reconnaissance platform, supporting rather than replacing human expertise.

4.3 Key Capabilities

The Robo Rescue Spider is designed around several capabilities that directly respond to the requirements identified in emergency-response robotics research.

1. Six-legged mobility

The robot uses six articulated legs to move across uneven or obstructed terrain. Each leg can be independently controlled, allowing the robot to coordinate its movement and maintain contact with the ground.

Research into emergency-response robots identifies mobility across different terrain types, obstacles and confined passageways as important robotic capabilities.

2. Environmental sensing

The robot can carry cameras and additional sensors to collect information about its surroundings.

The purpose of these sensors is to provide the operator with information that may be difficult or unsafe for a person to obtain directly.

3. Remote operation

The robot is designed to be controlled remotely. This allows the operator to remain at a safer distance while the robot investigates the environment.

NIST specifically identifies remote operation and safe standoff distances as important considerations for emergency-response robots.

4. Real-time visual feedback

A camera system can provide the operator with a live visual representation of the environment.

This can assist with navigation, obstacle identification and situational awareness.

5. Mathematical motion control

The robot uses mathematical modelling to coordinate its leg movements.

Forward kinematics can be used to determine the position of the foot from known joint angles, while inverse kinematics can determine the required joint angles for a desired foot position.

This provides the mathematical foundation for controlled and repeatable leg movement.

6. Gait control

The robot can use coordinated gait patterns to move its six legs efficiently.

The gait system determines when each leg enters the stance phase and swing phase, allowing the robot to move while maintaining an appropriate support configuration.

Research on hexapod robots has demonstrated the use of adaptive gait-control approaches for walking over irregular terrain while maintaining stability.

7. Modular sensing and payload capability

The platform can potentially be adapted to carry different sensors or payloads depending on the application.

For example, a search-and-rescue configuration could prioritise cameras and environmental sensors, while an industrial inspection configuration could use inspection-specific sensors.

This makes the concept potentially applicable beyond emergency rescue.

8. Simulation and digital validation

Because the project currently has limited access to physical hardware, the robot's movement and control systems can be developed and tested through simulation.

<u>Simulation allows the team to investigate:</u>

- ❖ Leg movement.
- ❖ Joint angles.
- ❖ Gait patterns.
- ❖ Foot trajectories.

- ❖ Stability.
- ❖ Obstacle negotiation.
- ❖ Control algorithms.

This also provides a practical way of validating the mathematical and software components before physical implementation.

4.4 Why a Hexapod?

The decision to use a hexapod configuration is based on the requirements of the problem rather than simply choosing six legs because the design looks like a spider.

A rescue robot may encounter uneven surfaces, obstacles and confined areas. NIST research identifies the ability of mobile robots to negotiate different terrain types, obstacles and confined passageways as important requirements for critical operations .

A six-legged configuration provides several potential advantages for this application.

1. Multiple points of ground contact

Six legs provide multiple points of contact with the ground. This allows the robot to distribute its body weight across several supporting legs.

The arrangement can provide a stable support configuration while other legs move.

2. Stability during movement

A hexapod can maintain several supporting legs on the ground while another group of legs moves.

For example, a tripod gait can divide the six legs into two groups of three. While one group provides support, the other group moves forward. The groups then alternate.

This creates a structured method of maintaining support during walking.

3. Ability to negotiate uneven terrain

Unlike a conventional wheeled platform, a legged robot does not require every part of its body to remain at the same height as the ground.

Individual legs can be positioned at different heights, allowing the robot to adapt its foot placement to variations in terrain.

Research on hexapod locomotion has demonstrated the use of controlled gait systems for movement over irregular terrain .

4. Obstacle negotiation

The legs can be individually positioned around or over obstacles.

This gives the robot additional freedom in choosing where to place its feet compared with a simple wheeled platform.

5. Redundancy

A six-legged configuration provides additional support points. If one leg encounters difficulty, the remaining legs can potentially provide support while the control system adjusts the robot's movement.

However, this does not mean that the robot can automatically operate normally with a failed leg. Actual fault tolerance would need to be experimentally tested and validated.

6. Suitable platform for mathematical modelling

The hexapod configuration also provides an appropriate platform for applying the mathematical and computational skills required by the project.

Each leg can be modelled using:

- Coordinate systems
- Forward kinematics
- Inverse kinematics
- Joint-angle calculations
- Trajectory planning
- Gait algorithms
- Stability analysis

This allows the project to combine mechanical design, mathematics, computer science and robotics into one integrated system.

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